Milestone #1: Project Proposal & Dataset Overview

Decoupling Economic Growth from Grid Emissions: Clean Power Transition Ahead of COP31

Alperen Efe Kuşçu

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1. Project Summary & Motivation (SDG 13 & COP31 Context)

As Türkiye prepares to host COP31 in Antalya, tripling global renewable energy capacity and phasing out fossil fuels remain central to achieving UN Sustainable Development Goal 13 (Climate Action). Because electricity generation is the single largest contributor to global greenhouse gas emissions, decarbonizing the power grid is critical for meeting national climate pledges. This project visually explores how major economies and Mediterranean nations have transformed their electricity mix between 2000 and 2025, examining whether economic growth (GDP) is effectively decoupling from grid carbon intensity. Ultimately, the resulting visual poster aims to foster public awareness ahead of COP31 by contrasting historical fossil fuel reliance with the rapid, seasonal, and regional rise of clean energy.

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2. Datasets & Source Information

This project merges two standardized open-access panel datasets (structured as 1 row per location and year) via **ISO-3 country codes (`iso_code`)** and **year (`2000–2025`)**:

1. **Ember — Yearly & Monthly Electricity Data (Primary Dataset):**
   - **Provider:** Ember (Independent global energy think tank, licensed under CC BY-SA 4.0).
   - **Content:** Annual and monthly electricity generation (TWh), percentage shares across 9 fuel sources (Solar, Wind, Hydro, Nuclear, Bioenergy, Coal, Gas, Other Fossil, and Other Renewables), electricity demand per capita, total power sector emissions ($\mathrm{MtCO_2}$), and grid carbon intensity ($\mathrm{gCO_2/kWh}$) for 215+ countries.
   - **Source URL:** https://ember-energy.org/data/yearly-electricity-data/ (https://ember-energy.org/data/yearly-electricity-data/)
2. **Data on $\mathrm{CO_2}$ and Greenhouse Gas Emissions by Our World in Data:**
   - **Provider & Authors:** Our World in Data — Pablo Rosado, Hannah Ritchie, Max Roser, Edouard Mathieu, and Bobbie Macdonald (Open access under Creative Commons BY license).
   - **Underlying Sources:** Compiled from the *Global Carbon Budget (Global Carbon Project)*, *Jones et al.* (national contributions to climate change), *University of Groningen GGDC's Maddison Project Database* (for GDP), *World Bank* (for country income groups), and *Energy Institute / U.S. EIA* (for primary energy consumption).
   - **Content:** Macroeconomic, demographic, and emissions indicators including real Gross Domestic Product (`gdp`), `population` , `co2_per_gdp` , and `primary_energy_consumption` .
   - **Source URL:** https://github.com/owid/co2-data (https://github.com/owid/co2-data)